African Leadership University

Bachelor of Software Engineering Program

Machine Learning Techniques II

Modeling Human Activity States Using Hidden Markov Models

Formative 1 Assignment

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GITHUB REPOSITORY:

https://github.com/pmugabo/Formative_1_Hidden_Markov_Models

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Human Activity Recognition using Smartphone Accelerometer and Gyroscope Data

1. Background and Motivation

Human activity recognition is important in smart health monitoring, fitness tracking, rehabilitation, and safety systems. In this project, the group use case is a smartphone-based fitness and safety monitor that can identify whether a user is standing, walking, jumping, or completely still. These activity states are not directly observed by the model; instead, they are inferred from noisy accelerometer and gyroscope measurements collected using a smartphone. A Hidden Markov Model is suitable because human movement occurs sequentially and the current activity is usually related to the previous activity.

2. Data Collection

Data was collected using the Sensor Logger mobile application on 1 July 2026. The sensors recorded were the accelerometer and gyroscope, each with x, y, and z axes. Four activities were recorded: standing, walking, jumping, and still. A separate unseen session was collected for model evaluation. The still activity was recorded by placing the phone on a flat surface, while standing was recorded with the phone held steady at waist level. Walking and jumping were recorded while performing the movements continuously.

Dataset	Activity	Sensor	Sampling Rate (Hz)
raw	jumping	Accelerometer	99.78
raw	jumping	Gyroscope	99.78
raw	standing	Accelerometer	99.78
raw	standing	Gyroscope	99.78
raw	still	Accelerometer	99.78
raw	still	Gyroscope	99.78
raw	walking	Accelerometer	99.78
raw	walking	Gyroscope	99.78
unseen	jumping	Accelerometer	99.78
unseen	jumping	Gyroscope	99.78
unseen	standing	Accelerometer	99.78
unseen	standing	Gyroscope	99.78
unseen	still	Accelerometer	99.78
unseen	still	Gyroscope	99.78
unseen	walking	Accelerometer	99.78
unseen	walking	Gyroscope	99.78

To harmonize recordings, all signals were resampled to a target sampling rate of 50 Hz before feature extraction. This ensured that feature windows contained comparable numbers of samples across all recordings.

3. Preprocessing and Feature Extraction

The accelerometer and gyroscope CSV files were loaded separately and merged using their seconds_elapsed timestamp. The data was cleaned by keeping the shared time interval between both sensors and resampling onto a uniform 50 Hz time grid. Features were extracted using 1-second windows with 50% overlap. Time-domain features included mean, standard deviation, variance, signal magnitude area, and correlations between accelerometer axes. Frequency-domain features were computed using the Fast Fourier Transform and included dominant frequency and spectral energy for each sensor axis.

4. HMM Setup and Implementation

The hidden states of the HMM were standing, walking, jumping, and still. The observations were feature vectors derived from the accelerometer and gyroscope windows. The transition matrix represented the probability of moving from one activity state to another, while the emission model used Gaussian distributions to describe the probability of observing a feature pattern given a hidden state. Initial probabilities, transition probabilities, means, and variances were initialized from the labelled training data and refined using Baum-Welch expectation maximization. The Viterbi algorithm was then used to decode the most likely activity sequence for the unseen recordings.

5. Results and Evaluation

The trained HMM was tested on unseen recordings collected in a separate session. The overall accuracy on the unseen data was 0.951. The confusion matrix and decoded activity plot show how closely the predicted sequence matched the true activity labels.

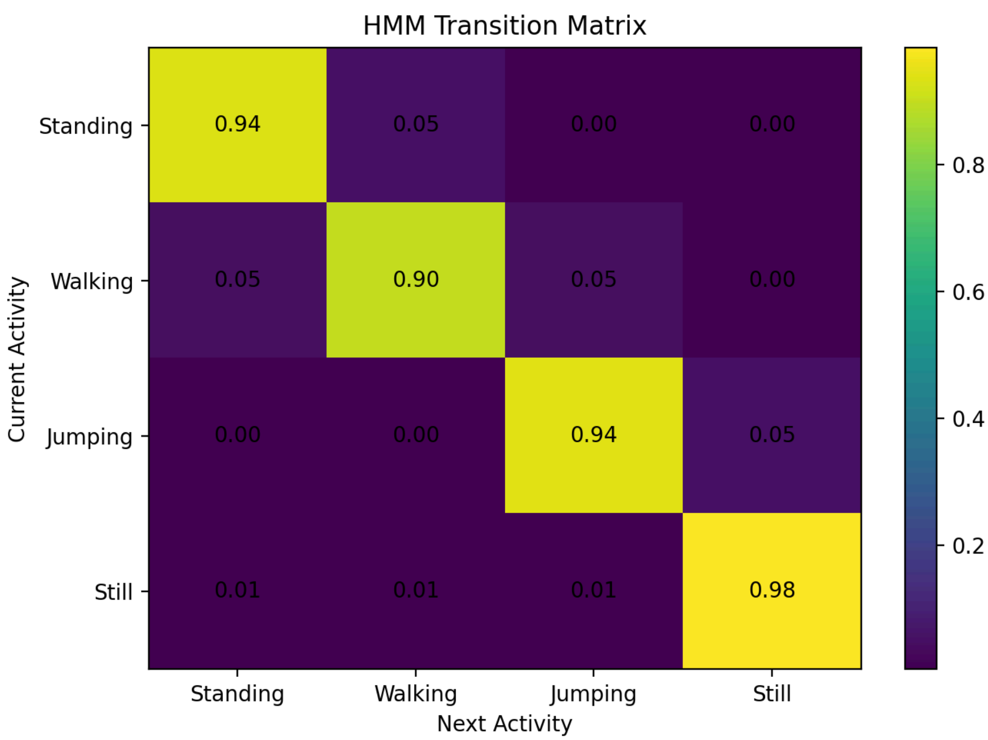

Figure 1. HMM transition matrix heatmap.

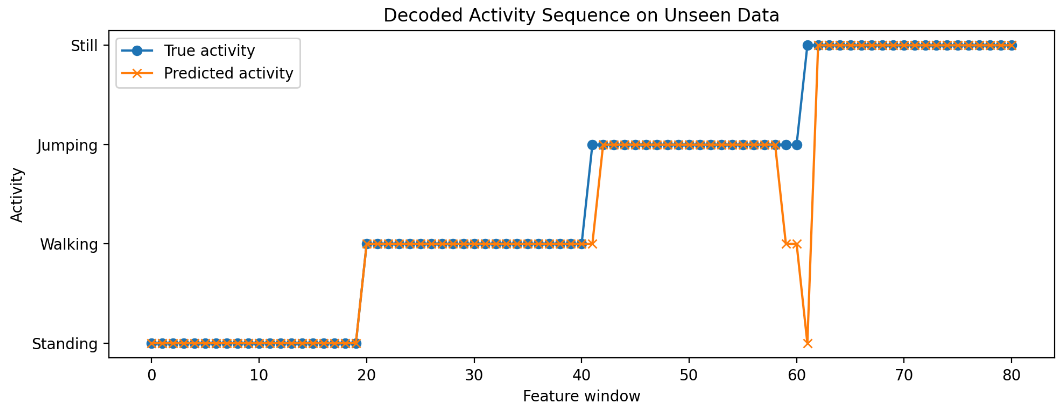

Figure 2. Viterbi decoded activity sequence on unseen data.

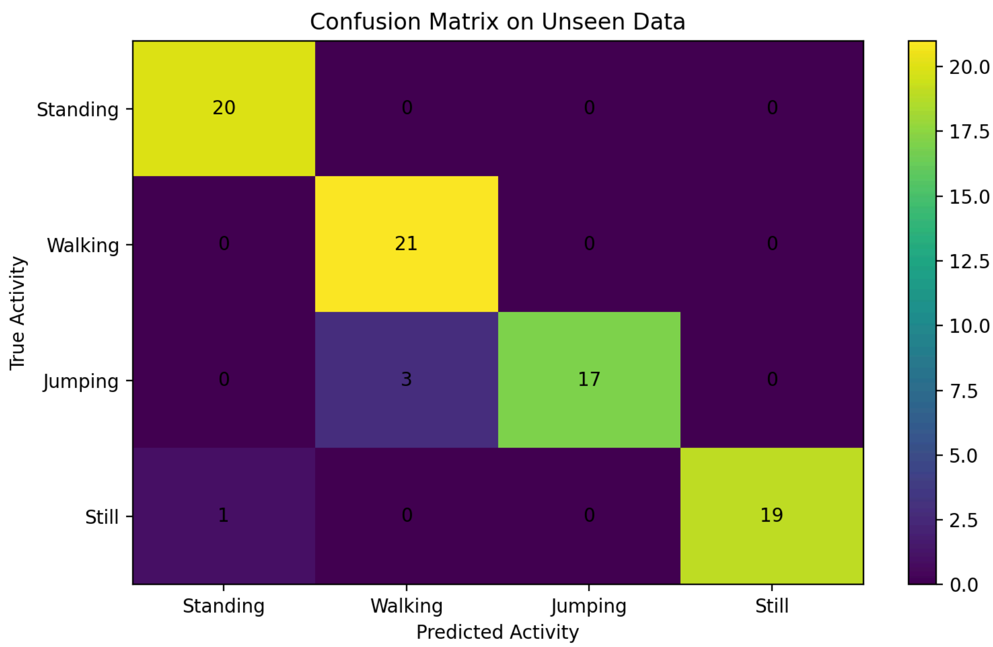

Figure 3. Confusion matrix for unseen data.

State (Activity)	Number of Samples	Sensitivity	Specificity	Overall Accuracy
Standing	20	1.0	0.984	0.951
Walking	21	1.0	0.95	0.951
Jumping	20	0.85	1.0	0.951
Still	20	0.95	1.0	0.951

6. Discussion

The easiest activities to distinguish were jumping and still. Jumping produced large accelerometer and gyroscope changes, while still produced very small signal variation. Standing and still were more similar because both involve limited movement, but standing still contains small hand and body movements that are not present when the phone is placed on a table. Walking was identifiable because it produced repeated rhythmic changes in the sensor signals. The transition probabilities reflected realistic activity behavior because the model assigned stronger probabilities to remaining in the same state across nearby windows. Sensor noise and small changes in how the phone was held affected the measurements, especially for standing and walking. More data from different participants, longer recordings, additional phone positions, and extra features could improve generalization.

7. Conclusion

This project successfully collected real smartphone motion data and used it to train a Hidden Markov Model for human activity recognition. The model used accelerometer and gyroscope features as observations and inferred the hidden activity states through Viterbi decoding. Evaluation on unseen data showed that the HMM can model sequential human activity patterns and distinguish between standing, walking, jumping, and still states. Future work should collect a larger dataset and test the model in more realistic environments.